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Institute of Medical
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Computational Methods to Reduce Bias in Binary Endpoints in Dose-finding Studies

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Alexandra Balzer
Institute of Medical Biometry
University of Heidelberg

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Introduction

- Optimization of dose finding and selection is a key goal in oncology drug research and development.
- Traditional approach: select maximum tolerated dose (MTD)
  - Problem: MTD may not reflect optimal efficacy and safety
- OPTIMUS Project: FDA initiative to improve dose optimization/dose finding
- Key challenge: Accurate estimation of therapeutic effects
  - True treatment effect is often overestimated in Phase II
  - Phase III results may not confirm Phase II findings.
- Current: Bias-correction methods were presented for normally distributed outcomes (e.g. bootstrap methods, [Zhan, 2024])
- No methods exists yet for binary endpoints as tumor response or toxicity.

Previous Work

Proposed Methods for normally distributed endpoints

In [Zhan, 2024] several methods are compared for reducing bias in normally distributed endpoints:

- Jackknife,
- **Single, Double and Triple (Non-) Parametric Bootstrap**,
- Shrinkage Estimator by Hwang [Hwang, 1993] and Lindley [Lindley, 2018],
- Hybrid estimators.

Setup

- I active treatment arms with n patients.
- Response X_{ij} for patient j in group i with $X_{ij} \sim g(\theta_i)$ and $i = 1, \dots, I$.
- $\mathbf{X_i} = (X_{i1}, \dots, X_{in})$ vector of responses for group i and $\mathbf{X} = (\mathbf{X_1}, \dots, \mathbf{X_I})$.
- The parameter of interest is $\theta_{max} = \max(\theta_1, \dots, \theta_I)$.
- Estimator: $\hat{\theta}(\mathbf{X}) = \max[\bar{\mathbf{X}}_1, \dots, \bar{\mathbf{X}}_I]$.
- Even if $\bar{\mathbf{X}}_i = \text{mean}(X_{i1}, \dots, X_{in})$ is an unbiased estimator, $\hat{\theta}(\mathbf{X})$ is biased.

Idea of the Bootstrap Method

- The bias of the initial estimator is:

$$\text{Bias} = \text{E}[\hat{\theta}(\mathbf{X})] - \theta_{max}.$$

- Bootstrap can be used to estimate this bias:

$$\widehat{\text{Bias}} = \hat{\text{E}}[\hat{\theta}(\mathbf{X_B})] - \hat{\theta}(\mathbf{X}).$$

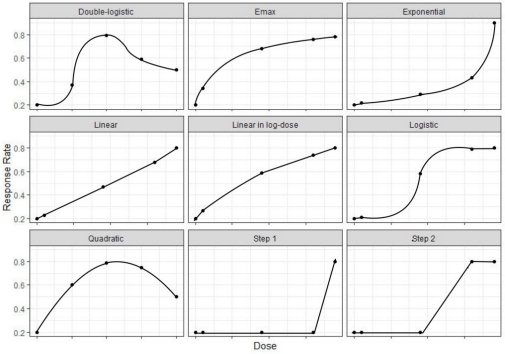
- This yields a bias-reduced estimator:

$$\hat{\theta}^{(1)}(\mathbf{X}) = \hat{\theta}(\mathbf{X}) - \widehat{\text{Bias}}.$$

My Simulation Study

General Setup

- The simulation study consists of three parts.
- For each part, 10.000 repetitions are performed.
- The study design includes nine dose-response relationships each with 5 doses [Ivanova et al., 2012].
- For each dose level, $n = 40$ patients per arm are considered.
- Binary endpoints are examined (e.g., response).



Part I - Optimization of Bootstrap Repetitions

Aim: Optimize the number of bootstrap repetitions B for the single and double (non-)parametric bootstrap in binary endpoints, based on bias and MSE.

- Triple bootstrap is not considered in this part due to computational burden.
- The number of bootstrap repetitions B is set to 80, 100, 200, 500, and 1000.

Part II - Bayesian Hierarchical Models

Aim: Identify the optimal hyperprior (with corresponding parameters) for a Bayesian hierarchical model to minimize bias and MSE.

The model is specified as: $Y_i \sim \text{Bin}(n, p_i), \quad i = 1, \dots, 5$ with

$$\text{logit}(p_i) \sim \mathcal{N}(\mu, \sigma^2), \quad \mu \sim \mathcal{N}(-1.38, 100).$$

- Hyperpriors determine how much information is pooled across groups (e.g., σ^2 controls degree of shrinkage).
- The following MCMC settings are applied fo all fits: burn-in 2.000, number of chains 3, total MCMC iterations per chain: 10.000.

Part II - Priors

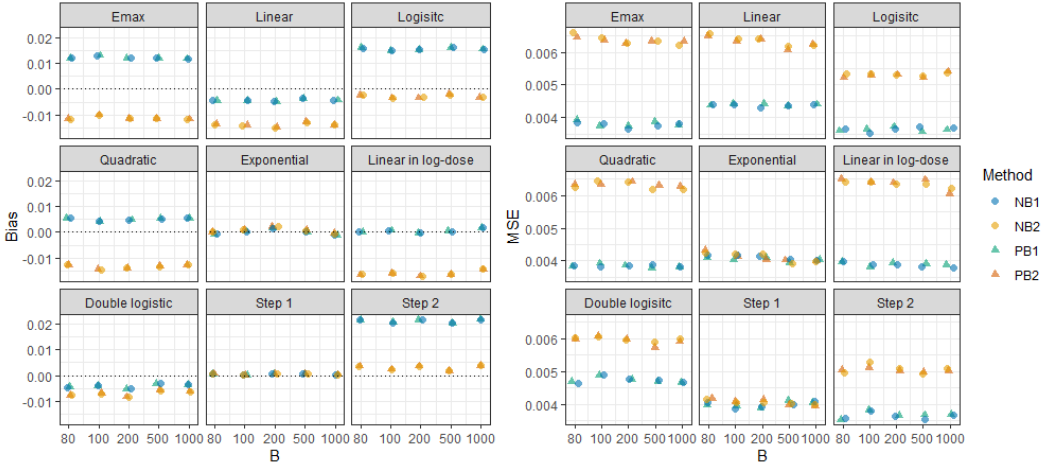
The priors and corresponding parameters to evaluate are based on [Cunanan et al., 2019]:

- Unif$[0, a]$ for $\log(\sigma), \sigma, \sigma^2$ with
  - $a \in \{1, 2, 5, 10, 25, 50, 100, 500, 1000\}$.
- Half-t for σ with
  - scale parameter $s \in \{0.5, 1, 2.5, 10, 25, 50, 100, 250, 500, 1000\}$ and
  - degrees of freedom $df \in \{1, 2, 5, 10, 25, 50, 100, 500\}$.
- Gamma distribution $\Gamma(a, b)$ for $\tau = \frac{1}{\sigma^2}$ with
  - $a \in \{0.005, 0.05, 0.25, 0.5, 1, 2, 5\}$, and
  - $b \in \{0.0005, 0.001, 0.005, 0.01, 0.1, 0.5, 1, 2, 5, 20, 50\}$.

Results

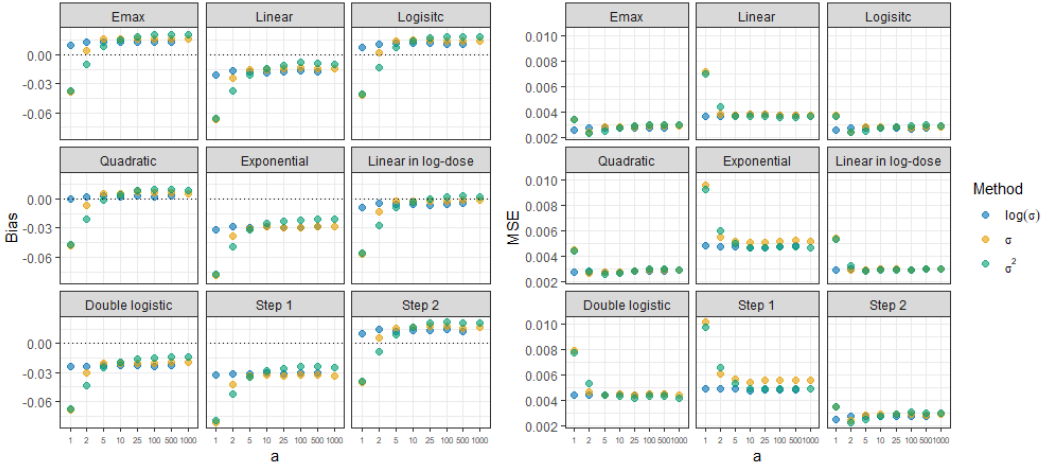
Part I

Part I


Emax
Linear
Logisitc
Quadratic
Exponential
Linear in log-dose
Double logistic
Step 1
Step 2
Bias
0.02
0.01
0.00
-0.01
MSE
0.006
0.005
0.004
80 100 200 500 1000
B
Method
NB1
NB2
PB1
PB2

Part II: Uniform

Part II: Uniform



Part II: Half-t

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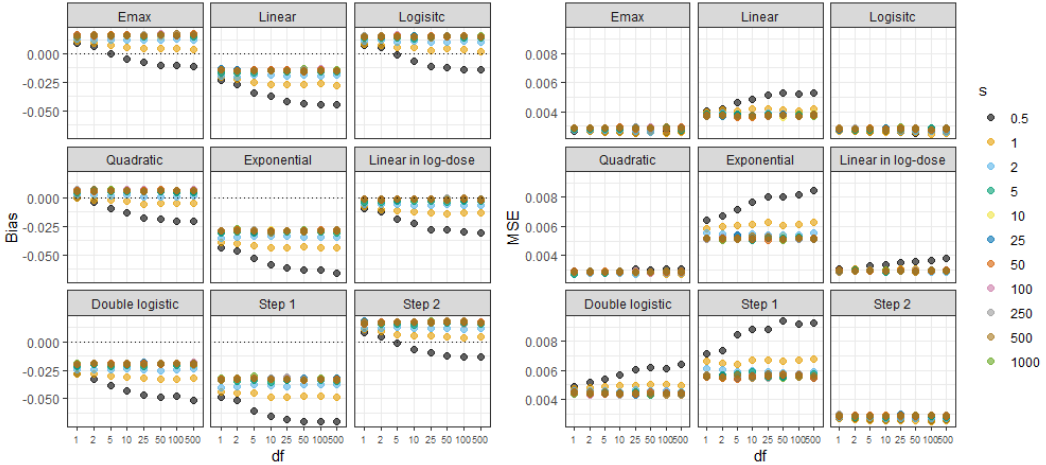

Part II: Gamma

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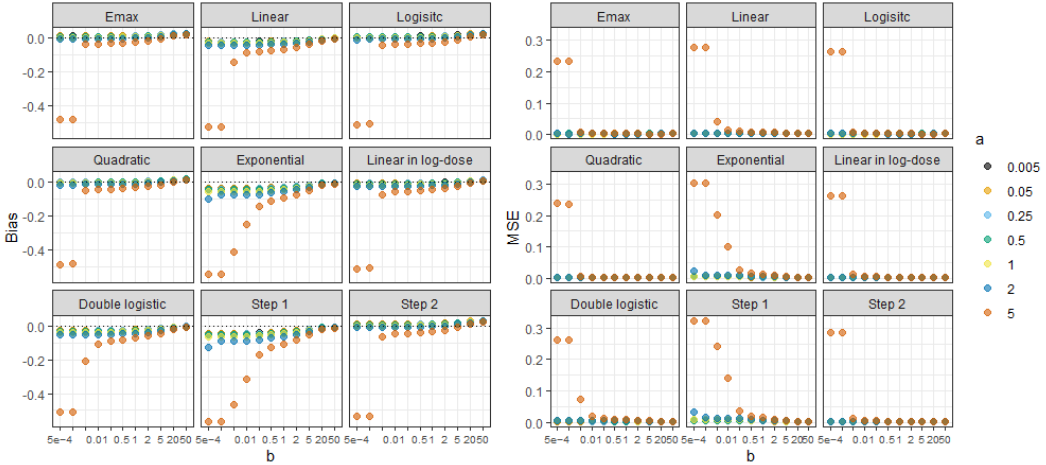
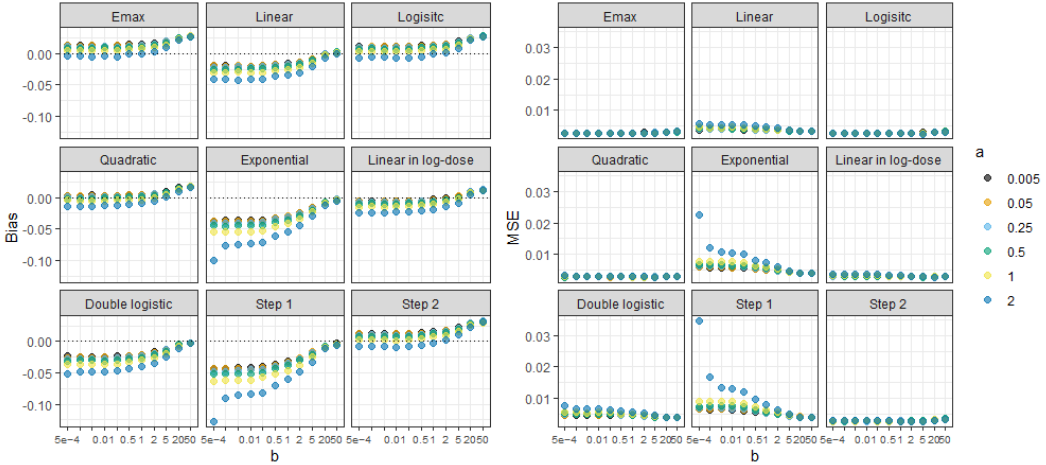


Part II: Gamma



Part II: Summary

Method	Parameter				Bias (MSE)
Uniform $\log(\sigma)$	$a =$	2			-0.0070 (0.00351)
Uniform σ	$a =$	100			-0.0047 (0.00371)
Uniform σ^2	$a =$	1000			0.0003 (0.00357)
Half-t	$df =$	100	s =	500	-0.0043 (0.00369)
Gamma	$a =$	0.25	b =	5	0.0004 (0.00335)

- Uniform prior on σ^2 with $a = 1000$ and Gamma prior with $a = 0.25$ and $b = 5$ show the smallest bias.
- MSE is very similar for most parameter values and hyperpiros, except of extreme outliers.

Further Considerations

- Simulations will also be performed with smaller groups sizes, e.g. n $= 20$; depending on the results, further sample sizes will be investigated.
- Triple Bootstrap will be analysed for $B = 80$; depending on computation time, additional runs with $B = 100$ and $B = 200$ may be considered.
- The maximal response in each dose-response relationship will be reduced to 60%, i.e. all responses will be scaled accordingly.

Part III

Based on the findings from Part I/II ($B = 80$, Gamma prior on $\tau = \frac{1}{\sigma^2}$ ($a = 0.25$, $b = 5$) and Uniform on σ^2 with $a = 1000$), the following will be considered for the next comparison:

- Additional methods (Jackknife, Shrinkage Estimator, Hybrid estimators) from [Zhan, 2024] will be simulated.
- For the Hybrid estimator [Zhan, 2024], Double Bootstrap and the Shrinkage Estimator were combined. Based on Part I, Single Bootstrap and Shrinkage Estimator may be combined.
- Methods proposed by [Kirby et al., 2012] (additive and/or multiplicative adjustment) will be applied.

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